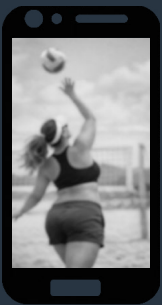
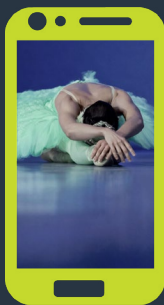
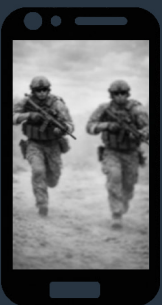
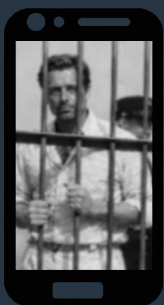
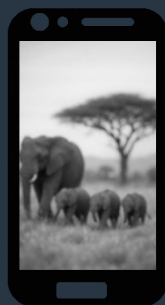
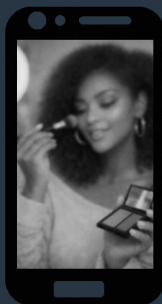
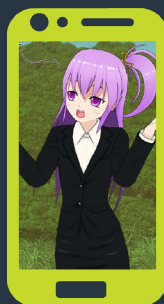
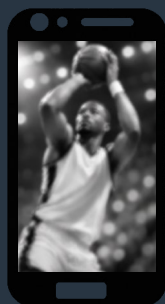


MOBILE PERSONALIZATION

The history and theory of
mobile streaming media



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MOBILE PERSONALIZATION

*The History and Theory of
Mobile Streaming Media*

by
Justin Grandinetti &
Charles Ecenbarger

Mobile Personalization: The History and Theory of Mobile Streaming Media

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CHAPTER ONE

The History of Mobile Personalization

On January 6, 2016, Netflix's co-founder and then-chief executive Reed Hastings delivered a keynote announcement to mark the platform's expanded global presence to more than 130 countries. Speaking to the Las Vegas Computer Electronics Show crowd, Hastings boasted:

Today you are witnessing the birth of a new global Internet TV network. With this launch, consumers around the world—from Singapore to St. Petersburg, from San Francisco to Sao Paulo—will be able to enjoy TV shows and movies simultaneously—no more waiting. With the help of the Internet, we are putting power in consumers' hands to watch whenever, wherever and on whatever device.¹

Within the year, Hastings continued the “Netflix-as-media-disrupter” narrative, this time targeting cinema. Commenting on a question about the future of cinema, Hastings juxtaposed the stagnant innovation of the movie industry with Netflix's desire to “unleash film” and grow the movie business by upending the traditional release model of films.²

Just a few years later, founder Jeffrey Katzenberg and CEO Meg Whitman of the aforementioned and much-maligned short-form streaming app Quibi similarly positioned the platform as a milestone in televisual evolution. Katzenberg made the following comments in a 2018 interview in which he attempted to generate buzz for Quibi:

¹ “Netflix Is Now Available Around the World,” *Netflix Newsroom*, Netflix, January 6, 2016.

² Janko Roettgers, “Netflix CEO Reed Hastings: Movie Theaters Haven't Innovated Beyond Popcorn,” *Variety*, March 17, 2017.

You take a TV in your pocket every day and ten minutes it happens all the time, waiting at the doctor's office and on your commute and it's taking longer and waiting for a friend for dinner, and we want to take that ten minutes and want to make those moments extraordinary by the service we're offering . . .³

In what ended up being an impressive publicity cycle relative to actual platform impact, the founder/CEO duo further emphasized that Quibi would also feature "cinematic content" broken down into quick, bite-sized portions for mobile streamers.⁴ Quibi, however, never took off, and the platform's content was eventually sold piecemeal to other services.

Aside from being streaming platforms, Netflix and Quibi aren't much alike. Netflix started as a snail mail delivery DVD rental service, while Quibi was born digital. Netflix operates globally and its name remains synonymous with streaming, whereas Quibi fizzled in under a year. And unlike Quibi, Netflix is not mobile specific (even though estimates show that the percent of mobile device access of the platform is growing each year). Nonetheless, what both of these platforms *do* have in common is a discursive connection with the televisual and cinematic. Put another way, there is clearly a link between our popular understanding of streaming today and audio-visual antecedent technologies—and major platform representatives have gone out of their way to cultivate this mindset.

As noted in the introduction, contemporary mobile streaming takes many forms, even when just limiting focus on the audio-visual. Not all of this content is televisual or cinematic in nature; for example, social media posts and mobile game streaming also complicate any attempt at a neat delineation. This conceptual morass leaves scholars of mobile streaming in a sticky situation. What do we make of statements by major platform representatives and our colloquial proclivities to situate mobile streaming as the latest step of a linear evolution of television? Or is streaming more akin to the movies—especially as prestige programming situates itself as cinematic? What about the social aspects of many mobile streaming platforms? And that's not to ignore mobile audio-visual game streaming . . .

It certainly makes logical sense that platform providers would invoke the popular and established media of television and cinema to facilitate the embrace of streaming. Television and cinema are familiar audio-visual de-

³ Jacob Wolinsky, "Katzenberg and Whitman Discuss Their New Streaming TV Venture, NewTV," *ValueWalk*, January 31, 2020.

⁴ Nicole Sperling, "What is Jeffrey Katzenberg's Quibi All About, and Why Should You Care?" *Vanity Fair*, June 14, 2019.

livery mechanisms that have stood the test of time and delivered countless hours of entertainment for mass audiences. Who wouldn't want to engage with the "future" of these familiar entertainment media?

And it also follows that there are some material similarities in terms of content and, increasingly, a convergence of delivery mechanisms. There are, of course, strands of media studies research that account for the seemingly inescapable televisual and cinematic lineage of streaming. As noted in the introduction, scholars have examined parallels and distinctions between television and new media, in turn rethinking the ontology and epistemology of the televisual itself.⁵ Put differently, to know the future of television one needs to understand what television *is* and *was* in the first place. There are others who focus on how streaming movies has heavily disrupted established cinematic release models—a phenomenon precipitated by the uneven recovery of cinema following Covid-19.⁶ Others still decenter the content of the televisual by instead examining the function of the screen itself as part of everyday activities and the structuring of space.⁷ Some even go so far as to claim that digital screens are best understood outside the paradigm of specific media; instead, film, televisions, computers, and telephones should no longer be treated as discrete.⁸ Contrary to the corporate bluster from streaming platform representatives, scholars of televisual, film, streaming, and screens do not merely accept that streaming is the direct and inevitable next form of television and cinema; nevertheless, these conceptual frameworks attempt to demystify the link between established mass media formats and contemporary streaming.

This book, however, is about *mobile* streaming. The history of the televisual is most often linked to the fixed site of the living room since the 1950s. Major film premiers have taken place in brick-and-mortar cinemas beginning in

⁵ Lynn Spigel, "Introduction," in *Television After TV: Essays on a Medium in Transition*, ed. Jan Olsson and Lynn Spigel (Duke University Press, 2004); Lisa Parks, "My Media Studies," *Television & New Media*, 10, no. 1 (2019); William Boddy, "Is It TV Yet? The Dislocated Screens of Television in a Mobile Digital Culture," in *Television as Digital Media*, ed. James Bennett and Niki Strange (Duke University Press, 2011); James Bennett, "Introduction: Television as Digital Media," in *Television as Digital Media*, ed. James Bennett and Niki Strange (Duke University Press, 2011).

⁶ Michael D. Smith and Rahul Telang, *Streaming, Sharing, Stealing: Big Data and the Future of Entertainment* (MIT Press, 2017), 34.

⁷ Anna McCarthy, *Ambient Television* (Duke University Press, 2001); Anna McCarthy, "From Screen to Site: Television's Material Culture, and Its Place," in *The Screen Media Reader: Culture, Theory, Practice*, ed. Stephen Monteiro (Bloomsbury, 2017), 389.

⁸ Kathleen Oswald and Jeremy Packer, "Flow and Mobile Media: Broadcast Fixity to Digital Fluidity," in *Communication Matters: Materialist Approaches to Media, Mobility, and Networks*, ed. Jeremy Packer and Stephen B. Crofts (Wiley (Routledge, 2012), 277.

the early 1900s. Yet, alongside the stationary imaginings of the audio-visual is an oft-ignored mobile and portable lineage. Our goal is to consider how mobile streaming has become so quotidian, embedded, and important. Doing so means taking seriously historical antecedent technologies to disrupt neat linear narratives. We believe that we must account for the persistent connection between streaming and television and film. But that doesn't mean that mobile streaming is merely the next evolution of these technological compositions. In essence, it's critical that we be cautious of industry hype cycles about the future of televisual and cinema via streaming as well as colloquial linear narratives of technological advancement from cinema to television to streaming to mobile streaming; at the same time, we also can't completely ignore, write-off, or disregard these persistent discursive connections between streaming and historical audio-visual antecedent technologies.

To make sense of this situation, we begin with the notion that mobile streaming is not merely a cause nor effect, but a complex and shifting mix of material technologies and discursive imaginings of audio-visual content on-the-go function toward mobile personalization. That is, how mobile and portable technologies are critical components in the ongoing transformation of times and spaces, by providing entertainment and information for users while simultaneously operating toward the goals of monetization and control.

In this chapter, we trace some mobile and portable antecedents to contemporary audio-visual mobile streaming: the portable television and the drive-in cinema. These technologies are often overlooked or on the margins of a more fixed and static history; nonetheless, they are critical to grasping how we arrived at the current moment of mobile and portable personalization. In what follows, we first detail this book's approach before examining each of these two technological antecedents, with particular attention to the monetization spaces and times through novel portable and mobile technologies.

Tracing Mobile Personalization

Our entry point for considering the historical lineage of mobile streaming comes from Raymond Williams, whose canonical work on the emergence and integration of the televisual into everyday life situates television as a cultural technology. In other words, Williams examines how culture and technology are intimately intertwined.

Media theory before Williams was often characterized by notions of technological causality and determinism. For instance, the Payne Fund Studies of the late 1920s through early 1930s, thought to be some of the earliest scholarship in the emerging field of media studies, used qualitative and quantitative social science methodologies to decipher the connection between juvenile delinquency and violent cinema. These studies set the tone for scholarly accounts designed to assess the consequences that the burgeoning mass media landscape had on individuals.⁹ Media effects research, born out of the new twentieth-century paradigm of mass media, often focuses on the relationship between “media” as content and how audiences make sense of what they see. Similar scholarship focusing on the relationship between content and societal effects has continued for decades.¹⁰ Relatedly, some strands of critical theory from the mid-twentieth century, though varied in how these theorists study the tête-à-tête between culture and technology, is similarly concerned with the impact of a powerful media on easily influenced populations. The Frankfurt School, for example, wrote in response to what they saw as an erosion of individual thought when the individual in a hyper-consumerist landscape internalizes mechanisms of social control.¹¹ Taken together, there is a near one-hundred-year-old proclivity to focus on media as content, to research how this content affects society (or vice versa), and to treat this relationship as relatively unidirectional.

By contrast, Marshall McLuhan is an enduring theorist whose writings eschewed attention to content to instead provide an expansive definition of media as mediator. His radical redefinition of “media” included clothes, money, and roads.¹² Instead of questioning how individuals interacted with mass media content, McLuhan focused on the scale, pace, and pattern new technologies introduced into human affairs. While McLuhan’s proto materialist accounts of technology provide an interesting and important direction for media studies, his work is tinged with a different kind of determinism—that resulting from his belief in technology’s capacity to increase individual freedom.¹³ As explained by Tiziana Terranova, McLuhan’s vein of techno optimism also characterized many discussions of new media in the

⁹ Herbert Blumer and Philip M. Hauser, *Movies, Delinquency, and Crime* (Macmillan, 1933).

¹⁰ Elizabeth M. Perse, *Media Effects and Society* (Laurence Erlbaum Associates, 2001), 1.

¹¹ Herbert Marcuse, *One-Dimensional Man: Studies in the Ideology of Advanced Industrial Society* (Beacon Press, 1964), 9.

¹² Marshall McLuhan, *Understanding Media: The Extensions of Man* (MIT Press, 1964).

¹³ James Bennett, “Introduction,” in *Television as Digital Media*, ed. James Bennett and Niki Strange (Duke University Press, 2011), 9.

early 2000s.¹⁴ Although McLuhan helped popularize an important avenue of ingress for media studies by going beyond content, his work is mired in determinist thinking.

Countering McLuhan, Raymond Williams rejects the notion that technology itself causes cultural change. Instead, Williams examines how the televisual emerged as a provisional outcome of complex social processes—in large part the increased mobility and privatization of American family life in the 1950s. By tracing the emergence of a particular technology as entangled with cultural structures, Williams provides a model for understanding how the emergence and acceptance of a technology cannot be divorced from its historical time period.

Of course, Williams's scholarship is not without some well-earned criticism of its own. Like many scholars of the time, Williams prioritizes a human-centric notion of agency. Media studies work has, over time, come to understand agency as relational, highlighting the role of technical machines as contingent components in larger societal assemblages.¹⁵ Whether it's assemblage theory, actor-network theory, object-oriented ontology, or vital materialism, media studies scholarship after the so-called materialist turn often highlights the importance of non-humans in expressing, maintaining, and transforming relationships of power.¹⁶ In other words, more materialist accounts strike a balance between the passive inevitability of McLuhan's technological determinism and Williams's overly human-centered and culturally driven focus.

One approach to assessing the intricacies of the relationship between culture and technology over time is what's known as media archaeology. Most generally, media archaeological techniques disrupt canonized narratives of media history to trace how the specific affordances, constraints, and discourses of technical media are constitutive of power relations in a given cultural

¹⁴ Tiziana Terranova, "Cyberculture and New Media," in *The Sage Handbook of Cultural Analysis*, ed. Tony Bennett and John Frow (Sage, 2008), 597.

¹⁵ Jennifer Daryl Slack and J. MacGregor Wise, *Culture and Technology: A Primer*, 2nd ed. (Peter Lang Publishing, 2015), 138.

¹⁶ Bruno Latour, *Science in Action: How to Follow Scientists and Engineers Through Society* (Harvard University Press, 1987); Donna Haraway, *A Manifesto for Cyborgs: Science, Technology, and Socialist Feminism in the 1980s* (Center for Social Research and Education, 1985); Gilles Deleuze and Felix Guattari, *A Thousand Plateaus: Capitalism and Schizophrenia* (University of Minnesota Press, 1980); Jane Bennett, *Vibrant Matter: A Political Ecology of Things* (Duke University Press, 2010); Gregory J. Seigworth and Melissa Gregg, "An Inventory of Shimmers," in *The Affect Theory Reader*, ed. Melissa Gregg and Gregory J. Seigworth (Duke University Press, 2010).

field.¹⁷ In other words, media archaeology is historically attuned in looking not for a linear evolution of specific technologies, but instead for moments of suppressed, neglected, and forgotten discursive and material manifestations of cultural technologies.¹⁸ As such, media archeological accounts are driven by a question of power: how clashes of competing technologies and discourses are posed as solutions to a given problem, eventually emerging as a temporarily stabilized sociotechnical apparatus.¹⁹

Media archaeology is often characterized by a focus on either discursive accounts indebted to Michel Foucault or more techno hardware materialist attention inspired by Fredrich Kittler. Foucault's archaeology is attentive to what he calls discursive formations. This approach looks beyond the discourse itself to understand how discourse functions as a locus of power. Take for instance this chapter's opening, which features statements by powerful industry executives about streaming as the future of television and cinema. These statements aren't the be-all and end-all final word on streaming; simultaneously, these discourses are powerful in shaping the colloquial understanding of how streaming fits into media history. By contrast, a materialist approach considers how technology changes over time, with specific focus on the dead ends, the forgotten, and the understudied. Televisual history, for example, is shaped by a competition between transmission mechanisms—the spinning mechanical disk versus the cathode ray tube.²⁰ Mechanical televisions were ultimately less successful and failed to reach market status. This oft-forgotten material struggle over format, function, and design is critical in understanding the development of the televisual.

Attention to the material *and* the discursive vis-à-vis media archaeology can go hand-in-hand. Technologies are more than materiality; they are a method of addressing problems systematically. That is, technical media arise out of distributed discursive networks (bodies of knowledge and techniques), but then intermix and recombine over time.²¹ Viewed from a media archaeological perspective, the history of mobile streaming can be

¹⁷ Erkki Huhtamo and Jussi Parikka, "Introduction: An Archaeology of Media Archaeology," in *Media Archaeology: Approaches, Applications, and Implications*, ed. Erkki Huhtamo and Jussi Parikka (University of California Press, 2011), 15.

¹⁸ Huhtamo and Parikka, "Introduction," 39.

¹⁹ Alexander Monea and Jeremy Packer, "Media Genealogy and the Politics of Archaeology," *International Journal of Communication* 10 (2016): 3150.

²⁰ Doron Galili, *Seeing by Electricity: Cinema, Moving Image Transmission, and the Emergence of Television, 1878–1939* (Duke University Press, 2020), 23.

²¹ Alexander Monea and Jeremy Packer, "Media Genealogy and the Politics of Archaeology," *International Journal of Communication* 10 (2016): 3147.

understood as part of continuities and discontinuities with both the fixed and mobile imaginings and material attempts at a mobile and portable televisual and cinema.

As a moment of brief recap and respite, our evaluation of mobile streaming is influenced by accounts of media that go beyond attention to content to examine how culture and technology are intertwined. Doing so requires threading a needle while walking a narrow tightrope. On one hand, we don't want to write off the persistent connection between the televisual, cinematic, and streaming. On the other hand, we want to challenge and shake up simple linear narratives of technological evolution that all-too-often fall into the trap of cultural and/or technological determinism.

Fortunately, we aren't blazing this academic trail on our own. We follow in the footsteps of recent scholarship that traces the cultural lineage of streaming technology, noting the long line of antecedent technologies as well as their accompanying cultural meanings and practices.²² Other accounts go beyond an electrical technology lineage to examine how streaming has roots in infrastructural and environmentally indebted technologies that attempted to control flow.²³ Our historical and mobile focus is inspired by Erkki Huhtamo's call for more attention to the mobile, portable, wearable, and nomadic when it comes to media archaeology.²⁴ Greater attention to the historical examples of mobile antecedent technologies has gained traction with mobile media scholars.²⁵ And in particular, our own examination of the history and development of mobile streaming intertwines with mobile historical scholarship that emphasizes the emergence of customizable media environments, portable personalization, and continuous consumption habits.²⁶

In the remainder of this chapter, we begin our historical inflection point with the context of the electrical audio-visual technological experiments of the turn of the twentieth century industrial modernity. Specifically, we

²² Burroughs, "A Cultural Lineage," 13.

²³ Grandinetti and Ingraham, "Infrastructures of Flow," 2.

²⁴ Erkki Huhtamo, "Pockets of Plenty: An Archaeology of Mobile Media," in *The Mobile Audience: Media Art and Mobile Technologies*, ed. Martin Rieser (Rodopi, 2011), 23.

²⁵ Meryl Alper, "Portables, Luggables, and Transportables: Historicizing the Imagined Affordances of Mobile Computing," *Mobile Media & Communication* 7, no. 3 (2018); Hugh Davies, "Fanology: Hand-fans in the Prehistory of Mobile Devices," *Mobile Media & Communication* 7, no. 3 (2019); Jordan Frith and Didem Özkul, "Mobile Media Beyond Mobile Phones," *Mobile Media & Communication* 7, no. 3 (2019); Mary Morley Cohen, "'Always on and Always on Them': Portable Radios and the Creation of the Mobile Media Experience," *Mobile Media & Communication* 4, no. 3 (2016).

²⁶ Morley Cohen, "'Always on and Always on Them,'" 371.

trace the understudied history of portable television and the mobility-linked cinema of the drive-in.

The Birth of Electrified Entertainment

Recent decrees of “the death of cinema” can make it easy to forget just how important and impactful the movie distribution model has been to mass communication for well over one hundred years. Movie theaters, much like the imagined site of the televisual, evoke a fixed and stable site. That is to say that cinemas are defined as much by the film and later digital projection technologies as they are by the structure of ritual practice these sites provide. People come to experience a copresence of what’s known as “one-to-many” audio-visual consumption.

It’s impossible to predict the future, but a betting person would likely take the position that the pinnacle of cinema is over. Cinematic spending peaked during the 1940s. Adjusted for inflation, box office rates have fallen year after year in the twenty-first century, with major hits to the traditional cinematic distribution model coming during and after Covid-19 lockdowns.²⁷ The popularity of cinema is, of course, dependent on how this technology is linked with societal events, other technologies, and changing distribution models. Even at a time when audiences seem more content than ever to wait for movies to come to streaming video so that they can enjoy them from the comfort of their own home, it’s important not to downplay cinema’s importance.

Recent innovations like “direct to streaming,” shorter theater release cycles, the demise of purchasing DVDs (and before that, VHS tapes) have fundamentally disrupted existing distribution models, and these changes are relevant to our contemporary understanding of what movies entail. In the era of mobile streaming, cinematic content can be accessed on-the-go via a variety of locations and mobile devices just as easily as it can be accessed from the couch at home. Cinema has, for much of its history, been viewed as a fixed audio-visual practice; but an examination of cinematic history reveals that portable and even mobile connotations have long existed alongside this more site-locked interpretation of cinema.

One such pre-cinema technology that complicates the notion of film as fixed is the magic lantern. The magic lantern is a relatively simple projec-

²⁷ “Domestic Movie Theatrical Market Summary 1995 to 2026,” *The Numbers*, <https://www.the-numbers.com/market/>.

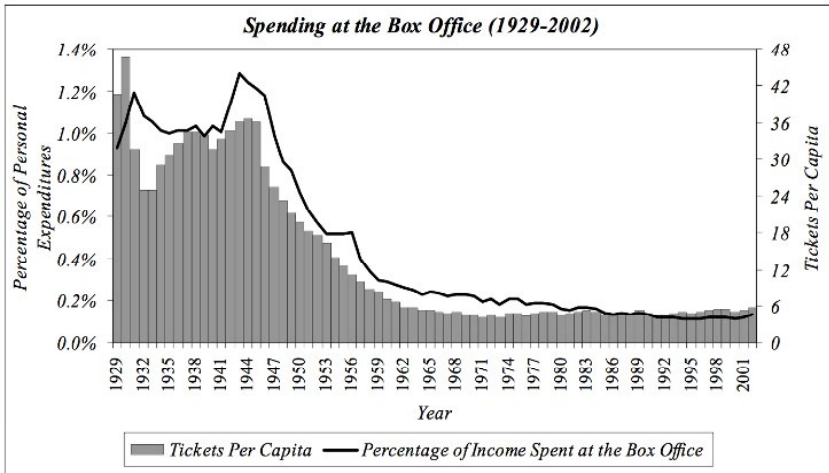


Figure 1.1. Spending at the box office (1929–2002). Orbach & Einay (2007, fig. 4).

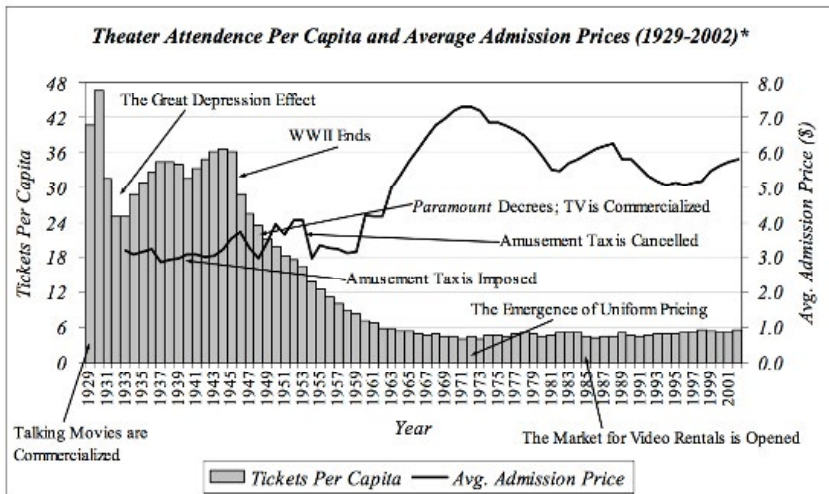


Figure 1.2. Theater attendance per capita and average admission prices (1929–2002). Orbach & Einay (2007, fig. 3).

tion technology, consisting of a box or chamber in which a light source is brought to focus to project light through an image painted onto glass.²⁸ The magic lantern was popular in eighteenth-century Europe, with the earliest known lantern show in America taking place in 1743.²⁹ Magic lantern shows

²⁸ Thomas Lamarre, "Magic Lantern, Dark Precursor of Animation," *Animation* 6, no. 2 (2011): 129.

²⁹ "A Brief History of the Magic Lantern," Magic Lantern Society, <https://magiclantern.org.uk/history/>.



Figure 1.3. The nineteenth-century magic lantern. Engraving courtesy of the Museum of London.

transformed a variety of spaces into sites of entertainment and education; by 1895, there were between 30,000 and 60,000 lantern showmen in the US, with an estimated 75,000 to 150,000 performances a year.

The magic lantern is just one of many burgeoning technologies of the nineteenth century that integrated a greater awareness of perception, namely the notion that stimuli can be divorced from their external referent, and that perception itself exists in time.³⁰ Stated differently, early visual technologies like the magic lantern and optical toys like the thaumatrope, stereoscope, and zoetrope marked the discovery that images can be linked together and the human mind will create a continuity of motion. Relevant to the history of cinema, these optical devices and techniques that were often initially devised for the purpose of scientific discovery were soon adapted into forms of popular entertainment. Furthermore, although devices like the magic lantern were part of a transformation of space into visual (with accompanying live audio) centers of entertainment and education, the technologies themselves are not fixed nor mobile but *portable*.

These precursors of visual entertainment set the stage for public fascination with the audio-visual. In the United States, a variety of inventors in the late 1800s patented the kinoscope (motion picture camera) and kintograph

³⁰ Jonathan Crary, *Techniques of the Observer: On Vision and Modernity in the Nineteenth Century* (MIT Press, 2001), 13.

(peephole viewing machine) in 1891.³¹ Early kinoscope exhibition was in many ways neither completely mobile nor fixed, as viewers watched short clips through the peephole of a large device. While inventor Thomas Edison patented the kinoscope, he did not initially ascribe substantial commercial value to the device.³² However, the kinoscope and early short films gained popularity, and the first kinoscope parlor opened on Broadway in New York City in April 1894, and the first European kinoscope parlor followed in October of the same year.³³ The kinoscope is often considered a direct precursor to cinema, but unlike the cinema structures that would soon take hold, these exhibitions did not always occupy a permanent space, as phonograph parlors, arcades, hotels, restaurants, etc. also attempted to capitalize on the novelty of the kinoscope by hosting the technology in a transient and temporary sense.

Chances are most people living today have never seen a kinoscope. That's largely because initial public fascination with the technology was quickly exhausted. That said, the kinoscope and other cinematic precursors are notable to our account of streaming media for two key reasons. First, interest in the technology established to inventors that there was a market for projection technologies. Second, these optical technologies make clear that cinematic precursors were not necessarily stationary and site-fixed media forms. While the cinema itself soon became housed in more permanent brick-and-mortar structures, the notion of cinema and its lineage as fixed in place is complicated by these optical experiments. It takes time to build infrastructure—in this case, the cinema structure itself. Simultaneously, the move from more portable to fixed is more than just preference; what the shift shows is that the site-specific evolution of early cinema was part of strategies of legitimation gained by tying the electric theater to burgeoning urban modernity. This evolution is critical to the connection between streaming and cinema, in that it becomes possible to view some of the historical antecedents to the mobile and portable audio-visual entertainment achieved by contemporary mobile streaming.

Rivalries and competing technologies were part of the continued construction of the Western cinema space in the coming decades. As public interest

³¹ Charles Musser, *Before the Nickelodeon: Edwin S. Porter and the Edison Manufacturing Company* (University of California Press, 1991), 30.

³² Norman O. Keim and David Marc, *Our Movie Houses: A History of Film & Cinematic Innovation in Central New York* (Syracuse University Press, 2008).

³³ Musser, *Before the Nickelodeon*, 44; Charles Musser, *The Emergence of Cinema: The American Screen to 1907* (University of California Press, 1994).

in early projection technologies took hold, peephole devices were replaced by storefront movie theaters dubbed nickelodeons.³⁴ These small spaces were often converted from the same arcades that exhibited the kinetoscope. Early displays often had a transient and mobile nature. For example, traveling projection exhibitors would often find an empty story space, set up in parks, or use the side of a downtown building.³⁵

Even though there is no inherent need for permanent structures for cinema exhibition, the cinema space soon became a fixture of urban modernity. Profitability and public interest in cinema were part of the construction of movie theaters, and New York theater-building reached its zenith by 1914.³⁶ Even so, popularity and profitability alone are not the only forces that led to the creation of movie theaters. An early discursive positioning of movies as “photoplays” gave the medium more legitimacy, akin to a night out at the theater.³⁷ Along with discourses that legitimated cinema, the materiality of audio-visual capture is important; movie projection can be and was done in a more temporary and transient form, but grandiose cinema spaces gave film an air of acceptability akin to that of a play. The fixed structure of the cinema precipitously took hold in the United States beginning at the turn of the twentieth century. Known early as “the electric theater,” the cinema became emblematic of the evolving urban spaces of modernity and industrialization in the Western world. In essence, there was (and perhaps still is) a power to building temples to cinema that one can trace back hundreds of years to pre-electrical theaters.

As mentioned at the start of this section, movie theaters still matter. According to the National Association of Theatre Owners, there were 39,798 indoor and 595 drive-in movie screens in the United States in 2017.³⁸ Simply stated, cinema spending per capita may have peaked in the 1940s, but the cinema structure endures as not only a Western but increasingly global fixture of urban spaces. While the first few decades of the twentieth century were dominated by the cinema, interest in the broadcast television, followed later by cable television, and eventually the internet created audio-visual competition to cinema. Yet, even in a more diverse landscape of media

³⁴ Keim and Marc, *Our Movie Houses*, 23.

³⁵ David Welling, *Cinema Houston: From Nickelodeon to Megaplex* (University of Texas Press, 2011), 10.

³⁶ Keim and Marc, *Our Movie Houses*, 27.

³⁷ Keim and Marc, *Our Movie Houses*, 25.

³⁸ NAC Editorial Team, National Association of Theatre Owners Report, 2024.

content, the ritual function of “going to the movies” persists alongside other fixed and mobile forms of entertainment.

At the Drive-in

Combining both cinema and automobility, the drive-in theater represents an alternate imagining of cinema. Unlike streaming from a smartphone, the drive-in cinema is not portable; the drive-in is, however, a conjunction of cinema, space, and mobility. Scholars of cinema consider the relationship between the materiality of the cinema and urban space largely because electrical projection technologies quickly became a fixture of urban space, dominating leisure time as part of expected movement through the city.³⁹ Throughout the past century, the cinema has, in turn, been part of urban modernity and industrialization, post-industrialization, and contemporary global relations. Much like the fixed spaces of urban cinema structures, the drive-in organizes movement to and from the cinema site itself. But the drive-in marks a departure from more traditional cinema spaces through its imbrication of mobile technology and how the drive-in offers a reconstruction not of urban, but instead *rural* and *suburban* spaces. As such, the drive-in represents an alternative construction of audio-visual entertainment that emphasizes mobility and a new engagement with space—a cinematic lineage relevant to mobile steaming.

Although commonly associated with the 1950s and 60s, the first drive-in theater opened on June 6, 1933, in Camden, New Jersey.⁴⁰ The idea for the drive-in is credited to Richard Hollingshead, who began his initial experimentation in 1928 using a Kodak projector, a sheet stretched between two trees, and a radio for sound.⁴¹ Hollingshead took inspiration from his mother’s discomfort sitting in traditional movie theaters, and was granted a patent for the drive-in theater in 1933.⁴² Though not as popular as the

³⁹ Bill Marshall, “Montreal Between Strangeness, Home, And Flow”; Adrian Fielder, “Poaching on Public Space: Urban Autonomous Zones”; J. Paul Narkunas, “Streetwalking in the Cinema of the City: Capital Flows Through Saigon”; Ronaldo B. Tolentino, “Cityscape: The Capital Infrastructuring and Technologization”; and Geoffrey Nowell-Smith, “Cities: Real and Imagined,” in *Cinema and the City: Film and Urban Societies in a Global Context*, ed. Mark Shiel and Tony Fitzmaurice (Blackwell, 2001).

⁴⁰ “First Drive-in Movie Theater Opens,” History.com, November 13, 2009.

⁴¹ Ben Ikenson, *Patents: Ingenious Inventions* (Black Dog & Leventhal, 2004), 241.

⁴² “First Drive-in Movie Theater.”

fixed site of the urban cinema, the number of drive-in theaters peaked in the United States at 4,063 in 1958.⁴³

In recreating the cinema experience, the drive-in captured the growing American interest in automobility. In describing this elision between the cinema and automobility, Hollingshead called the drive-in “a novel construction to outdoor theaters whereby the transportation facilities to and from the theater are made to constitute an element of seating facilities of the theater.”⁴⁴ To this end, the drive-in represents a relationship to space that was often quite different from its urban theater counterparts.

Because of the need to accommodate the space of automobiles, drive-in theaters required multiple acres of land. Most of the pre-1960s drive-in theaters were built on plots bigger than twelve acres, the largest of which was located on twenty-eight acres in Copiague, New York, with space for 2,500 cars and a full-service restaurant.⁴⁵ Since the drive-in was an open-air venue, these theaters had to be located far outside urban spaces and away from major highways. Consequently, the drive-in theater was part of an alternate idea of the cinema, instead structuring movement to and from rural spaces. Although perhaps not directly “mobile” during the motion picture projection itself, the drive-in theater does explain how cinematic technologies, along with a connection to automobility, could recreate not only cities, but other spaces as well.

The height of the drive-in’s popularity coincided with the rise of broadcast television and high rates of US auto ownership. Here again, Williams’s concept of mobile privatization is critical: Families became more mobile, increasingly moved to the emerging suburbs, and became more isolated. These social shifts necessitated new forms of contact outside the home, and these same needs were served via the drive-in theater. Drive-ins, at least initially, offered family-friendly entertainment and interconnection. In this regard, the drive-in theater represented a non-urban imagining of the theater space that directly implicated the automobility required to reach the theater into the construction of the theater space itself. This connection between the theater and the automobile maintained cultural underpinnings of mobile privatization—the family isolated within privatized space via their car as part of the theater.

⁴³ Steve Robrahn, “Drive-in Theaters Slowly Fade Out of Business,” *LA Times*, July 10, 1988.

⁴⁴ Ikenson, *Patents*, 241.

⁴⁵ Robrahn, “Drive-in Theaters.”

This family-centric construction of the drive-in space rapidly changed following the 1960s. As an outdoor venue, the drive-in is beholden to natural cycles of weather, temperature, and the requirement of the night sky to view movies in the first place. These limitations in scheduling were not lost on film distributors, who realized that drive-ins could only show so many movies per day. Soon, movie studios sent first-run releases to indoor theaters only. Drive-ins were left to show B-movies, and eventually, X-rated films.⁴⁶ This revision of the drive-in concept did allow some to survive, while simultaneously inscribing the mobile cinematic construction with youthful and “deviant” associations.

The decline in drive-in theater popularity was also precipitated by other elements. First, the relentless expansion of suburbanization rendered drive-in theaters unsustainable. As cities expanded outward, formerly cheap land became far too valuable to devote to a seasonal business model.⁴⁷ In addition, as the novelty of the family drive and automobility wore off, so did interest in the drive-in. One final contributor to the decline in drive-in theaters came in the construction of shared time. Daylight savings time went into effect in the United States in 1918, was repealed after World War I, and was subsequently reinstated during World War II. From 1973 to 1974, the impact of a fuel crisis on the United States led to a temporary national adoption of a year-round daylight savings time as a fuel-saving strategy.⁴⁸ One consequence of this additional hour of daylight was that drive-in showtimes were pushed later into the evening, and this further separated the drive-in space from the family-centric focus of previous decades. This confluence of global, local, economic, political, and environmental forces is accordingly part of the history of the drive-in cinema.

The drive-in theater offers a unique articulation between the cinema space, non-urban geography, and automobility. While scholarship on the early years of the cinema often focuses on the theater as part of an urban reconstruction of space, the drive-in offered a form of audio-visual entertainment adapted for areas considered on the periphery of modernization, industrialization, and electrification. Cinematic precursors certainly demonstrated that visual (with accompanying audio) entertainment could be set up almost anywhere. But it was the novel interest in automobility that

⁴⁶ Robin T. Reid, “The History of the Drive-in Movie Theater,” *Smithsonian Magazine*, May 27, 2008.

⁴⁷ Reid, “History of the Drive-in”; Robrahn, “Drive-in Theaters.”

⁴⁸ Paul B. Murray, “Did Daylight Savings Time Make the Drive-in Theater Go Dark?,” *Southern Outdoor Cinema*, March 31, 2012.

allowed drive-in cinema to gain traction as a mobile reconstruction of cinema space. This same desire to watch “anywhere” is directly relevant to the potentiality of mobile streaming, even if the technologies, infrastructures, and the content itself have changed. At the same time, factors like daylight savings time, changing distribution models, and urban/suburban expansion show that it is impossible to divorce technologies from their sociotechnical milieu. The fact that drive-in theaters have dwindled over time while the “traditional” cinema has thrived also shows the power of a fixed urban site.

Despite its decline in popularity, the drive-in offers a critical example of desire for an alternate, mobile orientation to cinema. The fixed site of early movie theaters was part of a new structuring of time and spaces in modern cities. The decades of rapid social and technological change that accompanied industrial modernity forever altered the patterns and pace of everyday life. Such changes were not limited to work hours; technologies of entertainment like the cinema changed the temporality and spatiality of leisure as well. And as shown by the example of the drive-in cinema, experimentations imbricated with the mobile extend and transform these new sociotechnical patterns to spaces outside urban life.

The Spaces of the Televisual

Just as Quibi founder Jeffrey Katzenberg positioned smartphones as a “TV in your pocket,” a 2017 AT&T Wireless commercial shows potential customers the possibilities of mobile streaming through a smartphone with the dialogue:

With AT&T you can get your entertainment right here, right now. When you get the incredible iPhone 7 from AT&T you can get unlimited data. And live TV. The channels you love. Your favorite shows and movies. Making your iPhone more of a . . . (iPhone ring tone and call screen interrupts the commercial flow) . . . oh, my TV is ringing.

The quip “My TV is ringing” is a joke about the multiplicitous uses and media convergence of mobile devices.⁴⁹ Just as importantly, this advertisement, while humorous in nature, reveals how the televisual and smartphone technologies have become increasingly intertwined. Mentioned in the introduction of this chapter, Netflix made the grandiose proclamation that its service was

⁴⁹ “Ringling,” AT&T Wireless TV advertisement, July 2017, https://www.ispot.tv/ad/w1_N/at-and-t-wireless-ringling.

the world’s first global television network. Due to discursive positionings such as these, it should be unsurprising that streaming distributors have been construed as part of a linear evolution of the televisual.

Television history itself is commonly divided into periods of TVI, TVII, and TVIII. The chart below shows the common temporal boundaries and general characterization of each. If one was to subscribe to such a rigid history, then it would naturally follow that streaming is a form of “TVIV.” Media theorists have nonetheless challenged such a position, noting that Netflix and other streaming providers might be more accurately considered a “matrix media” in which viewing patterns, branding strategies, industry structures, and media content interact and intermix in ways that shift away from the traditional television set.⁵⁰ Here, the materiality of media and use practices complicate established definitions and disrupt linear histories of the televisual. Or, put differently, it is important to recognize that film and cable, satellite, and broadcast television still co-exist with fixed and mobile streaming. Mobile streaming does not replace nor render these alternatives vestigial. Mobile streaming is more popular than ever, and although mobile streaming practices are not always strictly “televisual,” the connection between the two media forms cannot be ignored. Specifically, thorough examinations of televisual history emphasize the medium’s early domestic and fixed history. However, there are material and discursive precedents of mobile streaming that require greater inquiry.

TV Periods	Years	Characterization
TVI	mid 1950s to early 1980s	Channel scarcity, mass audience, three-network hegemony
TVII	early 1980s–late 1990s	Channel/network expansion, quality television, network branding strategies
TVIII	late 1990s–present	Proliferating digital platforms, further audience fragmentation, shift in commodity relations

Figure 1.4. Televisual periodizations. Adapted from Jenner (2014).

Locating the moment of origin for television is difficult, as, much like with the cinema, the televisual arose out of a convergence of invention

⁵⁰ Mareike Jenner, “Is This TVIV? On Netflix, TVIII and Binge-Watching,” *New Media & Society* 18, no. 2 (2014): 260.

and innovation across multiple decades. One potential starting point is the 1897 creation of the cathode ray tube, a specialized vacuum tube capable of transferring refracted electron beams into images.⁵¹ This technology is still utilized in televisions, video cameras, and other image technologies. In 1927, the television was patented by Philo Farnsworth, who later sold the rights to RCA for one million dollars.⁵² Alongside these material milestones, the emergence of the televisual is as much a story of how the technology entered popular imagination. As noted by theorists, the televisual follows a trajectory of technologies like the camera obscura, telescope, telephone, and telegraph that sought to connect two distant points in real time.⁵³ Key to the televisual, then, are notions of simultaneity and storage—especially in relation to optics. In essence, the televisual and the cinema became linked to one another. Television offered “live” moving picture entertainment, whereas the cinema became associated with stored content.

Over the coming decades, the concept of the televisual changed more than the technology itself did, with various political and economic forces driving different ideas about what television should be, all while maintaining a focus on live television’s ability to engage simultaneous presence. The television model was eventually dominated by one-to-one transmission—a center to periphery model of communication—focusing on either simultaneity or storage.⁵⁴ In addition to these histories, the televisual and broadcasting are intertwined with specific social needs. As previously explained, Raymond Williams’s mobile privatization is a period-specific theorization of the deep contradictions between the televisual and automobility, wherein centralized, public transmissions was tied together with privatized reception. As families became more mobile, they also became more private, and television was part of a sociotechnical response to the need for contact outside the privatized space of the home.

This deep connection between the televisual and the domestic spaces remained both the material reality and dominant discursive positioning of television practices for decades, despite network proliferation, expanding times of televisual interaction, and infrastructural advances. For example, Lynn Spigel similarly examines the televisual in conjunction with the con-

⁵¹ Ikenson, *Patents*, 62.

⁵² Ikenson, *Patents*, 271.

⁵³ William Uricchio, “Television’s First Seventy-Five Years: The Interpretive Flexibility of a Medium in Transition,” in *The Oxford Handbook of Film and Media Studies*, ed. Robert Koller (Oxford Academic, 2008), published online September 18, 2012.

⁵⁴ Uricchio, “Television’s First Seventy-Five Years.”

struction of public and private, noting the way that televisual technology provided new connections while collapsing prior distinctions between public and private spaces. With television, the home is no longer divorced from the external world: recreation, consumerism, the outside filtered in.⁵⁵ At the same time, there were attempts to create a mobile and portable television as early as the 1950s, indicating that the goal of a ubiquitous audio-visual relation to mass communication media is also fundamentally intertwined with televisual lineage.

This same desire is apparent in the mobile and portable cinematic examples, making it evident that contemporary mobile streaming was not as a sudden shift that occurred with the creation of smartphones but rather a longstanding goal of audio-visual media within capitalist landscapes. The example of the portable television—specifically, the discursive and commercial positionings of this technology—highlights this understudied aspect of television as an antecedent to contemporary mobile streaming, affirming the connection between portability, youth, dynamism, futurism, and a reconstruction of the world into spaces of the televisual.

Mobile, Portable, and Televisual

Discursive imaginings and material experiments in mobile and portable television have existed for decades. The portable television, like many other technologies, has military roots. The electronic camera was the first video technology to become portable, with the camcorder and videorecorder following soon after.⁵⁶ Strapped to soldiers and TV cameraman, these portable technologies created a mobile orientation to television—the notion that TV could go anywhere.

In the 1950s, the US Army, Navy, and the National Defense Research Committee contracted RCA to improve upon wartime small and lightweight airborne television systems.⁵⁷ The idea of these portable devices was to create a television system that could pick up and transmit enemy movements from the front line to command centers. Of course, the production technologies

⁵⁵ Lynn Spigel, *Make Room for TV: Television and the Family Ideal in Postwar America* (University of Chicago Press, 2004), 11.

⁵⁶ Ricardo Cedeño Montaña, *Portable Moving Images: A Media History of Storage Formats* (De Gruyter, 2017), 80.

⁵⁷ Cedeño Montaña, *Portable Moving Images*, 137.

of television as mobile differs from the commercial technologies of home and portable television sets.

From a commercial perspective, the black and white picture portable era for television began in 1952, with the release of the first battery-operated transistorized television for sale in 1960.⁵⁸ While television manufacturers such as Sentinel produced what could be considered “portable” televisions due to their size (for example, the 1948 Sentinel 400TV 7”), the reliance on electric outlets limited the actual mobile uses of such models. Preceding Sony’s TV-8-301 model, television console sizes began to shrink. Following Sony’s success and public interest, myriad portable models were produced throughout the 1970s and 1980s.

Advertisements for various models of portable television provide insight into some of the popular discursive positionings and imaginings of the televisual untethered from the space of the living room. For example, Sony’s 1976 advertisement for a 13” portable model depicts a nuclear family watching television while lying together in bed (fig. 1.5). Although still a domestic image of the televisual, the portable nature of the television allows a reconstruction of home media engagement beyond the space (and normalized temporal schedule) of the living room.

The 1968 AWA advertisement depicted in figure 1.8 similarly highlights shifting family relations, imploring consumers to “be a two TV family” for “twice the fun,” as “today’s living demands” portability. Here, the living room televisual scene is juxtaposed with a bikini-clad young woman in a more exotic locale. The connection between youth and portability is again emphasized in General Electric’s advertisement, in which a son tells his father that their fixed television, and by extension living room space, is “all yours” (fig. 1.6).

Collectively, these advertisements position portability as associated with youth, dynamism, and the possibility of the televisual beyond the living room. Other advertisements, such as Sony’s 1960s promotions, engage more directly with the reconfiguration of space via the portable television. In this specific example, Sony tells customers that they can take the television with them to the beach, yacht, or cottage, or use the portability to watch television in the kitchen, bedroom, park, or garden (fig. 1.7). These selections are just some of many commercial discursive positionings of portable television; however, they display explicit engagement by corporate interests to sell the idea of a new dynamic and mobile televisual.

⁵⁸ “Television History: A Timeline 1878–2005,” Tarlton Law Library, University of Texas at Austin, <https://tarlton.law.utexas.edu/mason-and-associates/timeline>.

**NOW
A TUMMY TELEVISION
THE WHOLE FAMILY CAN
GET A BELLYFUL OF**



My how we've grown!
Time was when we made a Tummy Television
barely big enough for one person to watch.
Now we have a black-and-white portable that's all of 13"
(screen measured diagonally). And since it's 100% solid state,
has a glare-free screen for indoor/outdoor viewing
and an energy-saving system that shuts the power off completely
when the set is not in use, we expect the biggest problem you'll ever have
is agreeing on which program to watch.

**INTRODUCING OUR 13" PORTABLE
"IT'S A SONY!"**

©1976 Sony Corp. of America. SONY is a trademark of Sony Corp. Model TV-131 Black and White Portable. TV picture simulated.

Figure 1.5. Advertisement for Sony Model TV-131 Portable TV Tummy Television, 1977.

Much like the father-son ad example of figure 1.6, other advertisements connect portable television to futurism. Sony's dramatic call for individuals to "Hold the future in your hand" via portable television claims that the portable model is the "personal set predicted for the family of the seventies" (fig. 1.9). In addition to these bold claims, Sony notes some of the technical specifications and advances that make the model possible. Of course, the

brief text ends with a return to the forthcoming technological future, telling readers that someday all televisions will look like Sony's micro model, so why wait? One final example, yet again from Sony, characterizes their portable model as "personal" television (fig. 1.10). This discursive move is one alluded to in some of the previous examples—that portable television allows greater individuality, choice, and freedom. At the same time, the explicit engagement with personalization as synonymous with portability is a powerful imagining that has been revisited via contemporary mobile streaming practices.

It's all yours, Dad... I'll take the General Electric Portable

No wonder this is TV's most popular portable... it weighs only 26 pounds, performs like a console!

Just now a General Electric Portable can add to fun at your home—

Now you can enjoy your favorite programs while the kids see theirs. And you can sit and watch anywhere you choose to be. For this set—aptly named the "Companion"—goes where you go, from room to room, outdoors in summer, even away on trips and vacations. You see, it weighs only 26 pounds. Yet, wherever you plug it in, it gives a

clear, sharp picture. That's because it's equipped with an enormous picture tube, a dark safety window which helps to deepen contrast, and a built-in antenna. Just like high-priced models.

So start thinking about this new idea in television, and see your G-E dealer soon.

Ask about other sizes, too—the "Big-Screen" which weighs only 32 pounds, and G-E's handy new "Personal"—weighs less

than 12½ pounds, costs as little as \$69.95.

There are 3 handsome color combinations to choose from—take your pick. But HURRY—the Christmas rush'll soon be on General Electric Company, Television Receiver Dept., Syracuse 6, N. Y.

Progress Is Our Most Important Product

GENERAL ELECTRIC

Figure 1.6. Advertisement for General Electric's portable televisions, 1956.



9" TV BBC 1, BBC 2 and ITV. 43" telescopic aerial. Mains or battery operated (battery optional extra), 85 Gns.

THE TV THAT'S REALLY PORTABLE!

In just twenty years the Sony Corporation has become the world's leading name in electronics. Sony pioneered miniature transistorised radios and televisions. And the result of many years of research is shown above. It's the TV that's really portable. Take it with you: on the beach, the yacht, to the cottage. Use it in the kitchen, bedroom, park or garden; wherever you are take a Sony.

Sony make a complete range of transistorised radios and tape recorders. Full details from your Sony dealer or write for a colour brochure to: SONY U.K. SALES DIVISION, Dept. T1, Mercia Road, Gloucester. Tel: Gloucester 28841. London Showroom, 70-71 Welbeck Street, London, W.1. Tel: HUNter 2143



SONY®
RESEARCH MAKES THE DIFFERENCE

AP1268-B. Printed in UK

Figure 1.7. Advertisement for Sony portable TV, 1960.

Though these advertisements provide only a small selection of how corporations positioned the portable television, the relation of portability to dynamism, freedom, personalization, and new spaces of televisual engagement are commonplace throughout the 1950s, 1960s, and 1970s. Taken as a totality, these advertisements shed light on the connection between the televisual and the reconstruction of space, mobility, and personalization.

Consequently, it becomes possible to view mobile streaming media not as a wholly novel technological interplay, but one rooted in historical continuities and disruptions.

Be a two TV family - IT'S TWICE THE FUN!

Solve TV programme arguments in the family and have twice the fun from your TV entertainment... be a two TV family with an AWA "Marconi" Portable! Only AWA gives you the superb quality of Deep Image picture, rich natural sound, and a choice of screen sizes... 11", 19", 21" and now the completely new 17" Teletar Deep Image P4!

The "Teletar" glimmers with space-age technology. A sound system that features the new integrated circuit as used in space vehicles, plus a sound deflection panel for bigger full-front sound. Fingertip-smooth channel changing. Superb 17" Deep Image picture. Twin telescopic aerials. Carrying handle. AC operated. In Woodgrain Blue, Shell White.

AWA 19" Stereo Portable P2 - full front sound, up-front controls, twin telescopic aerials, private earpiece AC operated. Blue-Gal, Shell Blue, Maple or Teak Woodgrain.

AWA 21" "Master" P3 - full-front sound, top-front tuning, twin telescopic aerials. Ready grids in for easy carrying. Woodgrain Blue with simulated Rosewood Walnut Woodgrain.

AWA 11" Lightweight Portable "The Maroon" P1 - a "space-age" portable with big-art performance and full-front sound. Australia's lightest-weight portable! Features twin telescopic aerials, private earpiece for personal listening. AC operated. In teak-look Blue, teak-look Red or Shell White.

Today's living demands AWA DEEP IMAGE PORTABLE TV

Manufactured and marketed by AMALGAMATED WIRELESS (AUSTRALASIA) LIMITED, Australia's national Wireless Organisation
 Price \$2

THE AUSTRALIAN WOMEN'S WEEKLY - November 27, 1968

Figure 1.8. "Be a two TV family." Advertisement for AWA televisions, 1968.

The decline in popularity of portable television sets occurred long before the advent of smartphones and mobile streaming. The television was able to achieve rapid household penetration in large part due to its limited infrastructural requirements; existing electrical power coupled with analog

hold the future in your hand
with **SONY**
RESEARCH MAKES THE DIFFERENCE

THIS IS TELEVISION OF THE FUTURE. This is the personal set predicted for the decade of the Seventies. So light and compact you carry it with you like a book, wherever you go. Put it beside your bed, on your desk at the office, outdoors for picnicking on the patio, in the back of the car or on the boat. It plays anywhere on its own rechargeable battery pack, auto battery or AC, with a picture so bright and sharp ordinary sets pale by comparison. Weighing only 8 lbs., it is hardly larger than a telephone, yet it out-

performs standard receivers in sensitivity and durability. Available only in limited quantities, SONY brings it to you today through its advanced research in the epitaxial transistor, so powerful and sensitive it is used only in computers and other advanced electronic equipment—and the new Micro-TV. It would be no exaggeration to say that someday all TV will look like SONY Micro-TV. But why wait for someday? See it today at selected dealers. SONY Micro-TV list \$229.95. Optional battery pack.

See and hear the world famous SONY all-transistor, battery operated radios at selected dealers.
SONY CORPORATION OF AMERICA 514 Broadway, New York 12, N.Y.
In Canada: General Distributors Ltd. 701 Notre Dame Ave., Winnipeg

Figure 1.9. “Hold the future in your hand.” Advertisement for Sony televisions, 1962.

broadcast meant that most American households owned a television by the end of the 1950s.⁵⁹ However, early analog broadcast television, otherwise

⁵⁹ Woojin Kim, “Television and American Consumerism,” *Journal of Public Economics* 208 (2022): 3.

known as TVI, is marked by channel scarcity and three-network hegemony. This limited channel structure continued until the early 1980s, when cable television infrastructure allowed growth in both the number of channels and picture quality itself. The infrastructural limitations for portable television further situated the portable version of the television as inferior, a sentiment still often echoed today.

The tepid response to the portable television reveals the complex nature of technological dissemination. For example, the videophone, another audio-visual experiment that can be considered part of both the televisual and telephonic lineages, also failed to gain traction. Despite its conceptualization as early as the 1920s, as well as pre- and post-war experiments, including by Bell Laboratories, the expensive nature of service and infrastructural requirements halted the videophone.⁶⁰ The videophone is not a mobile technology; nevertheless, like the portable television, the videophone shows how technological development is itself a complex composition of often competing forces. By contrast, the portable radio saw commercial success beginning in 1939, with manufacturing beginning decades earlier, the imaginings of which began as an idea in science fiction narratives in 1890.⁶¹ These examples show how the success (or lack thereof) of technologies is deeply connected to their historical context. Returning, then, to the instance of the portable television, infrastructural and material limitations meant that the “personalization” of space through mobile televisual engagement was not yet possible. By extension, television continued for decades as a predominately fixed technology.

Television, historically connected to the space of the living room, has in the decades since its inception expanded beyond the domestic sphere. Scholarship on the televisual has positioned the television as connected to liveness, storage, and simultaneity, while screen studies often investigate the place of the material screen as part of a structuring of space.⁶² As it was clear from the advertisements analyzed above, mobile experiments

⁶⁰ Matt Novak, “A Brief History of the Videophone That Almost Was,” *Tech News* (blog), Gizmodo, August 28, 2013.

⁶¹ Michael Brian Schiffer, *The Portable Radio in American Life* (University of Arizona Press, 1991), 5.

⁶² Packer and Oswald, “Flow and Mobile Media,”; Paul Frosh, “The Face of Television,” in *The Screen Media Reader: Culture, Theory, Practice*, ed. Stephen Monteiro (Bloomsbury, 2017); Ethan Tussey, “Connected Viewing on the Second Screen: The Limitations of the Living Room,” in *Connected Viewing: Selling, Streaming and Sharing Media in the Digital Age*, ed. Jennifer Holt and Kevin Sanson (Routledge, 2014); Anna McCarthy, *Ambient Television* (Duke University Press, 2001).



SONY Now Offers a Choice In Personal Television

Regardless of the model you select, you choose the finest when you buy SONY personal, truly portable television. Whether it's the remarkable 8-lb. Micro-TV with 5" screen or the new 12-lb. Model 9-304W with 9" screen, it operates on its own rechargeable battery pack, auto/best power and AC. Designed by SONY, developer of the transistorized pocketable radio and leader in the field of transistor electronics, the 27-transistor Model 9-304W introduces innovations which make it

unique. Included are a new rectangular picture tube with sensitivity and resolution superior to ordinary sets (a special accessory filter provides unsurpassed resolution even in bright daylight); new Mesa transistors for increased sensitivity; automatic gain control for better picture stability; 1/10th the power consumption of ordinary sets and practically no heat. A full series of accessories, including battery pack and UHF adapter, is available for both models.

SONY
RESEARCH MAKES THE DIFFERENCE

SONY CORPORATION OF AMERICA • 380 FIFTH AVE., N.Y. 26, N.Y. • REGIONAL OFFICES: LOS ANGELES • CHICAGO

Figure 1.10. "A choice in personal television." Advertisement for Sony televisions, 1964.

of the television were untethered from the infrastructural constraints of domestic life. Portable television highlights the mobile-fixed dialectic also present at various points in film history. While the living room at the time of television's rise may have been a space best understood through Williams's mobile privatization, the portable television reflected a desire for the transformation of space. This portability and mobility were discursively linked to notions of futurism—specifically the transformation of any space into one of personalized convenience. With regard to portable television, what matters is that there have long existed mobile experiments in the televisual as a deterritorialization of space and mobile capture of desire.

As previously mentioned, Williams's mobile privatization provides a conceptualization of how the televisual encouraged a way of life that was both mobile and home-centered; however, his notion is specific to a TVI paradigm. In engaging with the work of Williams, Lynn Spigel offers "privatized mobility" to understand how miniature and portable televisions could be used to keep the family and domesticity together even while people were on the go.⁶³ Spigel's conceptualization is a crucial model for understanding the history of the televisual; whereas Williams studied the emergence and lived experience of the televisual, Spigel instead highlights the possibilities and fantasies of the televisual as unplugged from the home. Established by the brief selection of portable television advertisements, discursive positionings of the portable television do emphasize freedom, mobility, dynamism, etc., yet simultaneously, retain a focus on the family unit and a mobile form of domesticity.

Although this portable and mobile relation to the televisual was not fully achieved through early attempts at the portable television, contemporary mobile streaming forms reengage the desire for mobile personalization and recreation of spaces beyond the domestic context of traditional television. The roots of a personalized and dynamic television experience trace back to the early years of the televisual, made apparent by the portable television. While promoted as empowering for viewers, the portable television served an economic purpose by capturing new times and spaces that might otherwise have been television free. Though these devices were limited in popularity, these examples yet again demonstrate the lineage of mobile personalization.

⁶³ Lynn Spigel, *Welcome to the Dreamhouse: Popular Media and Postwar Suburbs* (Duke University Press, 2001); Spigel, *Make Room for TV*, 75.

The Present and Future of Mobile Personalization

The goal of this chapter was not to argue that the drive-in cinema and portable television are direct antecedents falling along a neat evolution from early optical technologies all the way to mobile streaming. What we do wish to highlight is an alternative and oft-forgotten mobile and spatial history of the audio-visual. The drive-in and portable television are, in fact, different both materially and discursively from contemporary web-based mobile streaming content. Yet, despite these differences, we believe that these moderately successful historical examples do in fact inform our current moment of mobile streaming.

Although cinema began structuring urban spaces at the turn of the twentieth century, there are alternate origins to mobile streaming that did not receive as much attention. The example of drive-in cinema explains an alternative imagining that engages directly with mobility. In turn, this example disrupts the myopic linear narrative of cinema as fixed by providing historical ruptures and disjunctures that offer the possibility of a different relation to audio-visual practices.

Perhaps most importantly, this mobile-fixed dialectic highlights how the mobile and portable has often been subordinated over time. Conceivably in large part due to the rhetoric of the electrical sublime, as well as modernization, industrialization, and urbanization, the fixed space of the cinema gained legitimacy and primacy that was not always afforded to mobile forms. Cinema is also part of the rise of the mass audience, while mobile alternatives often catered to those not part of an urban modernity and the accompanying homogenization of twentieth-century mass media. The space of the cinema cannot be divorced from its historical period, and it should come as no surprise that the electrical novelty of cinema became enshrined in urban spaces through brick-and-mortar structures. In this way, the cinema created a spatial relation that structured movement to, from, and around urban space.

Cinema as fixed structure offered legitimacy via the immobility of the media form itself. The cinema space has long served as an arbiter of cultural forms; the movie theater is the first place of presentation for serious film productions, whereas lesser movie productions are relegated to “straight-to-home” formats. This same hierarchy often continues today, through the standard release order of theaters followed by airlines and hotels, home video, video-on-demand and pay per view, premium TV, and finally broadcast TV. Of

course, this model has been disrupted heavily by streaming providers, which in turn challenges not only strategies of monetization, but also legitimation.

The drive-in reveals that mobile capture of a desire for entertainment has been part of a cinema lineage that predates the movie theater itself. Material limitations notwithstanding, the fixed space of the cinema has been the gatekeeper of film since the turn of the twentieth century—a part of modernist regimes of spatial construction and control. It is the articulation of film to the internet that resurrects some of the same mobile desires of cinema lineage. Understanding mobile engagement with film via streaming media requires attention to this spatial, discursive, and material dialectic between fixed and mobile.

This dialectic is also apparent via the example of the portable television. For much of televisual history, domestic spaces defined the practice of watching television. Scholars such as Anna McCarthy have provided thorough examinations of the proliferation of screens as part of the construction of space and power relations, yet this fixed and site-specific nature of audio-visual media is just one vision that often coexisted with mobile alternatives.⁶⁴ From the televisual perspective, the portable television offered a vision of broadcasting free from the electrical infrastructure and fixed space of domesticity. Analyzed through this historical perspective, it becomes possible to view mobile streaming not as an entirely new media form, but instead as part of both continuities and discontinuities of discursive and material mobile technological forms that have existed for decades.

The examples of the drive-in and portable television reveal imaginings of cinema and the televisual that are quite different from the contemporary complexity of mobile streaming. Simultaneously, what unites these media forms is the function of recreating space as a capture of desire—what we call mobile personalization. As established in this chapter, such a capture of desire has precedents in terms of the cinematic and televisual. Perhaps the greatest departure of mobile streaming comes via the form of digitization. It is no secret that streaming providers rely on data collection, storage, and processing—elements that are not inherent in quite the same fashion to cinematic and televisual technologies. As such, mobile streaming is part of a lineage of film and television, but it also reanimates the trajectory of early media like census, clocks, and calendars, wherein the purpose of storage, data, and processing has once again become emphasized through the digital.⁶⁵ It

⁶⁴ McCarthy, *Ambient Television*.

⁶⁵ Jeremy Packer, “Epistemology Not Ideology OR Why We Need New Germans,” *Communication and Critical/Cultural Studies* 10, no. 2–3 (2013); John Durham Peters, *The Marvel-*

is precisely this digitalization and data collection that allows mobile streaming to become a hyper-efficient form of the capture of desire—and space and geographic data are part of this collection and subsequent algorithmic recommendation.

This chapter traced the origins of mobile personalization, with roots in the early optical technologies, experiments with the electrical and cinematic, the intertwining of emergent automobility and the cinema, as well as the promised dynamism and youth of the portable television. Our mobile personalization approach considers these sociotechnical changes not merely as linear evolution, but as moments that help trace the lineage of a mobile and portable desire to transform time and space through technologies.

In the next chapter, we examine the architectures of mobility that facilitate contemporary mobile streaming. Here, we go beyond the smartphone to also consider the infrastructures and platforms that enable mobile streaming media. This includes the role of platform models, wireless infrastructures, and how platforms and infrastructures of mobile streaming have come to merge. Moreover, we consider some of the key technical underpinnings of mobile streaming, like big data, algorithmic models, and APIs. Smartphones are certainly the point of access when it comes to streaming content, but infrastructures of data flows also play a pivotal role in creating, sending, and receiving streaming media.